University of Computer Studies (Thaton)

2025-2026 Academic Year

Fifth Year (B.C.Sc.)

Advanced Artificial Intelligence II (CS-5112)

Project Proposal

No.	Contents	Details Description
1	Project Title	MyanLitBot: An AI-Powered Literary Assistant for Myanmar Fiction
2	Project Type	Group Project [Maximum: 4 students]
3	Problem Statement	Classic Myanmar literary works embody rich thematic depth, emotional nuance and cultural significance. However, students often face difficulties in conducting effective literary analysis without access to expert interpretation. Conventional keyword-based search systems are limited to surface-level text matching and fail to provide coherent, structured explanations suitable for academic study. While recent advances in large language models (LLMs) have enabled conversational interaction with text, generic chatbot systems frequently generate unfaithful or hallucinated content, particularly when applied to low-resource languages such as Burmese (Myanmar). Furthermore, most existing Burmese NLP systems primarily focus on question-answering tasks and do not adequately support paragraph-based, descriptive literary explanation, which is essential for academic literature studies. The Burmese language specifically suffers from a lack of large, structured digital corpora, unlike English which benefits from resources such as the NLTK Brown Corpus and Gutenberg Corpus. This scarcity significantly limits academic research, educational applications and literary analysis in Burmese. To address these challenges, this project proposes MyanLitBot—a Retrieval-Augmented Generation (RAG) system designed specifically for Myanmar novel analysis. By integrating semantic retrieval using sentence embeddings with an instruction-tuned large language model, the system produces paragraph-level literary

		explanations strictly grounded in the original novel text. This approach minimizes hallucination, ensures factual consistency and supports formal academic interpretation for educational and research purposes while contributing to the preservation and accessibility of Burmese cultural heritage.
4	Objectives	The objectives of this project are: ▪ To develop an intuitive and robust platform for exploring and understanding Burmese literary works ▪ To enable precise retrieval of the most relevant story content, supporting detailed study and analysis ▪ To generate insightful, human-like literary interpretations that facilitate critical understanding of narratives, themes and characters ▪ To support education and literary analysis in the Burmese language ▪ To demonstrate the potential of AI technologies in overcoming the challenges of low-resource language processing
5	Scope of the Project	**Included Scope** ▪ Multi-novel literary analysis with isolated indexing per novel ▪ Myanmar novel text processing and linguistic normalization ▪ Sentence-aware text chunking with overlap for context preservation ▪ Semantic retrieval using multilingual sentence embeddings and FAISS vector indexing ▪ Two-stage retrieval with cosine similarity reranking for improved accuracy ▪ Descriptive text generation using Gemma-3-27B-IT with strict prompt engineering ▪ Web-based chatbot interface with Gradio supporting multiple novel selection ▪ Persistent index storage system for zero re-processing on system restart **Excluded Scope** ▪ Speech, image or multimodal input / output ▪ Real-time collaborative features

6	Selected Topic Area	▪ Text Processing ▪ Information Retrieval ▪ Neural Language Models ▪ Large Language Models (LLMs) ▪ Retrieval-Augmented Generation (RAG) ▪ Semantic Search and Vector Databases
7	Dataset / Corpus Description	The dataset used in this project consists of **five classic Myanmar novels** spanning multiple genres: **1.** အမှတ်တရ (ဂျူး) – Genre: Love/Romance **2.** ပြိုးချိ်လဲကန်တော့ခံတော်မူပါ ရယ်ချိ်လဲကန်တော့ခံတော်မူပါ (နုနုရည်အင်းဝ) – Genre: Social Realism **3.** ဗိုလ်ချုပ်အောင်ဆန်း အတ္ထုပ္ပတ္တိ (သခင်ကိုယ်တော်မှိုင်း) – Genre: Biography **4.** သူလိုလူ (ဂျာနယ်ကျော်မမလေး) – Genre: Biography **5.** လက်ကျန်လရောင် (ပုညခင်) – Genre: Love/Romance Testing Corpus 1. သူငယ်ချင်းလိုပဲဆက်ချိ်ခေါ်မည်ခိုင်(တက္ကသိုလ်ဘုန်းနိုင်) 2. ဓားတောင်ကိုကျော်ချိ်မီးပင်လယ်ကိုဖြတ်မည်(မြသန်းတင့်) 3. မောင့်ဘဝညအလားကွယ်(တက္ကသိုလ်ဘုန်းနိုင်) **Language:** Burmese (Myanmar) **Format:** Plain text (.txt) files **Storage:** Google Drive with persistent processed data directory
8	Text Preprocessing Steps	The following preprocessing steps are applied to prepare the novel text for semantic retrieval and generation: ▪ Removal of excessive newlines and tabs ▪ Text normalization and whitespace standardization ▪ Sentence-aware segmentation using Burmese punctuation (။, !, ?) ▪ Overlapping chunk generation with sentence overlap to preserve context continuity ▪ Maximum chunk size of 500 characters with configurable overlap
9	NLP Techniques Used	▪ Sentence Embeddings and Semantic Similarity Search ▪ Retrieval-Augmented Generation (RAG)

		▪ Controlled Text Generation with Hallucination Prevention ▪ IVF Clustering for Efficient Semantic Search ▪ Cosine Similarity Reranking ▪ Output Deduplication and Cleaning
10	Algorithms/ Models Applied	▪ **LaBSE (Language-agnostic BERT Sentence Embedding) –** 768-dimensional multilingual sentence embeddings for semantic representation of Myanmar language ▪ **FAISS (Facebook AI Similarity Search) –** IVF (Inverted File Index) with dynamic cluster adjustment for high-performance vector similarity search ▪ **Gemma-3-27B-IT –** Instruction-tuned large language model accessed via API for text generation (zero local GPU memory usage) ▪ **Scikit-learn** – Cosine similarity computation for retrieval reranking
11	Tools & Libraries	Python – Core programming language Sentence-Transformers – LaBSE embedding generation FAISS – Vector similarity search engine Google Generative AI (google-genai) – Gemma-3-27B-IT Scikit-learn – Analytical utilities and cosine similarity Gradio – Web-based user interface Pickle – Persistent storage of sentence chunks Google Colab + GPU – Model execution environment
12	System Architecture / Workflow	The system follows the workflow below: **Phase 1: Data Preprocessing & Indexing (One-time)** ▪ Load Myanmar novel text from Drive. ▪ Clean and normalize text. ▪ Perform sentence-aware chunking with overlap. ▪ Generate LaBSE embeddings for each chunk. ▪ Create FAISS IVF index with dynamic nlist calculation. (max(10, min(100, len(embeddings)//5)) ▪ Save index (.index) and chunks (.pkl) persistently to Google Drive.

		Phase 2: Query Processing (Runtime) ▪ User selects novel and enters query topic. ▪ Encode query into embedding using LaBSE. ▪ Perform IVF cluster search in novel-specific FAISS index. ▪ Retrieve candidate chunks (top_k × 2 for broad search). ▪ Rerank candidates using cosine similarity (sklearn). ▪ Select top 10 most semantically relevant chunks. **Phase 3: Generation** ▪ Build constrained descriptive prompt with strict semantic boundary rules. ▪ Generate explanation using Gemma-3-27B-IT . ▪ Apply output cleaning (duplicate removal, incomplete sentence filtering). ▪ Display response with response time via Gradio UI.
13	Training & Testing Strategy	No supervised training is conducted in this project. Both the embedding model (LaBSE) and the large language model (Gemma-3-27B-IT) are used in inference mode with API access. **Testing Strategy:** ▪ System testing performed using multiple literary topics across all three novels. ▪ Test categories include: character , conflict , theme, setting and style , etc. ▪ Generated outputs manually reviewed to assess: Factual consistency with source text , Contextual grounding (no hallucination) , Academic suitability and formal Burmese language quality , Response coherence and completeness , Retrieval accuracy evaluated through relevance of returned chunks .
14	Evaluation Metrics	▪ **Retrieval Quality**: Relevance of retrieved chunks to query topic ▪ **Textual Faithfulness**: Absence of hallucinated content not found in source novel ▪ **Response Coherence**: Logical flow and narrative continuity ▪ **Output Cleanliness**: No duplicate sentences, complete sentences only

		▪ **Response Time**: End-to-end query processing duration tracking ▪ **Fallback Accuracy**: Correct triggering of "ဝတ္ထုအတွင်း မတွေ့ပါ" when topic not in novel
15	Expected Output	▪ Descriptive literary explanations in formal Burmese ▪ Strictly grounded in the original novel text with zero hallucination ▪ Multi-novel support with novel-specific retrieval ▪ Response time tracking for each query ▪ Clean, publication-quality text without duplicates or incomplete sentences ▪ Academic-style literary interpretation suitable for educational use
16	Practical Component	▪ The project includes a fully functional web-based prototype (Gradio interface) that allows users to: ▪ Select from multiple Myanmar novels via dropdown menu ▪ Input literary topics in Burmese language ▪ Receive context-grounded explanatory paragraphs generated by the system ▪ View response time for each query ▪ Access persistent storage ensuring zero reprocessing on system restart
17	Project Schedule	Jan 20 [Project Proposal Submission] Jan 27 [First Seminar – Progress Test] Feb 3 [Second Seminar – Progress Test] Feb 17-18 [Final Seminar]
18	Limitations	▪ The system currently supports a limited number of Burmese novels (5), restricted by available digitized text resources ▪ Retrieval accuracy depends on the quality of sentence chunking and embeddings; subtle literary nuances may still be missed ▪ The system focuses on descriptive literary analysis and does not yet support comparative or cross-novel reasoning
19	Expected Outcomes from Project	▪ Advanced understanding of NLP pipelines and LLM-based RAG systems

		▪ Practical experience with Burmese language AI processing ▪ Contribution to preservation and accessibility of Burmese cultural heritage ▪ Demonstration of AI potential for low-resource language processing
20	References	▪ Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W. and Riedel, S. (2020) Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Advances in Neural Information Processing Systems (NeurIPS 2020)*. ▪ Feng, F., Yang, Y., Cer, D., Arivazhagan, N. and Wang, W. (2022) Language-agnostic BERT Sentence Embedding. *Transactions of the Association for Computational Linguistics*, 10, pp. 56–72. ▪ Johnson, J., Douze, M. and Jégou, H. (2019) Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3), pp. 535–547. ▪ Pedregosa, F. et al. (2011) Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, pp. 2825–2830. ▪ Reimers, N. and Gurevych, I. (2020) Making Sentence Embeddings Useful for Semantic Search. *Proceedings of the 12th Language Resources and Evaluation Conference (LREC)*, pp. 43–49.